Compute a Reusable Spatial MEM Basis

Description

Computes exact or low-rank Moran eigenvectors and retains explicit state required to project the same basis to held-out locations.

Usage

```
compute_spatial_mev_basis(
  data_coords_projected = NULL,
  n_mev = 20L,
  strategy = "exact",
  exact_max_locations = 1999L,
  fast_eigenvectors = 200L,
  fast_seed = 900723L,
  exact_function = sjSDM::generateSpatialEV,
  fast_function = NULL
)
```

Arguments

`data_coords_projected`	Data frame with 'coord_x_km' and 'coord_y_km' columns and unique row names.
`n_mev`	Positive integer giving the number of public eigenvectors requested.
`strategy`, `exact_max_locations`, `fast_eigenvectors`	Shared strategy settings validated by [resolve_spatial_mev_strategy()].
`fast_seed`	Positive integer used locally for deterministic Nyström knot selection.
`exact_function`	Exact Moran-eigenvector function. Defaults to [sjSDM::generateSpatialEV()].

`fast_function`	Fast Moran-eigenvector function. Defaults to [spmoran::meigen_f()].

Details

Exact results retain the current sjSDM values unchanged. Fast results use an exponential-kernel basis and orient each selected eigenvector deterministically. The orientation is retained for matching projection. Fewer than four projected input rows is a registered data-feasibility outcome with observed and required counts, not an engine failure.

Value

Named list with 'data_mev', 'projection_basis', and one-row 'data_provenance'.